Assignment One

Requirements： (1) "To deal with plagiarism, a zero mark will be given if your copy someone else's work or you let someone copy your work."

(2) "Every assignment/project submission must be attached with a signed 'Declaration of Use of Generative AI'

Literature Searching Assignment: Developing a Multi-Database Search Strategy Based on PICOS

1. Assignment Title

The Effectiveness of Artificial Intelligence–Powered Conversational Agents in Supporting Self-Management and Improving Glycemic Control Among Adults With Type 2 Diabetes: A Multi-Database Search for Randomized Controlled Trials

2. Background

Type 2 diabetes is a chronic condition that requires continuous self-management, including blood glucose monitoring, medication use, dietary management, physical activity, problem-solving, and communication with healthcare professionals. Nurses play an important role in diabetes education, behavioral support, follow-up, and care coordination.

Artificial intelligence–powered conversational agents—including chatbots, virtual health assistants, embodied conversational agents, and voice assistants—can provide automated education, personalized feedback, reminders, behavioral support, and medication-management assistance. These technologies may supplement nurse-led diabetes management by delivering continuous support outside clinical settings.

Randomized studies have evaluated conversational agents for diabetes self-management and voice-based AI systems for insulin management. However, the interventions, outcomes, and terminology used in this field vary considerably, making a structured and reproducible literature-search strategy necessary.

3. Aim of the Assignment

The purpose of this assignment is to develop, conduct, document, and critically evaluate a reproducible multi-database literature search concerning AI-powered conversational interventions for adults with type 2 diabetes.

Students are expected to:

1. Translate a clinical nursing question into a PICOS framework. **[5 marks]**

2. Identify appropriate controlled vocabulary and free-text terms, and adapt the search strategy to different database interfaces. **[2 marks]**
3. Conduct both sensitive and specific searches, document the complete search process using screenshots. **[5 marks]**
4. Describe your search results. **[1 mark]**

Note: The following databases should be searched:

1. PubMed
2. Cochrane Library
3. China National Knowledge Infrastructure—CNKI
4. CINAHL
5. Web of Science Core Collection

For each database, students must record:

- Database and platform
- Search date
- Complete search strategy
- Limits applied
- Number of results
- Number of potentially relevant studies